

canoLiq

Liquid Staking Protocol on Canopy Network

Technical Whitepaper — Version 1.2

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Abstract

canoLiq is a liquid staking protocol built as a Nested Chain on Canopy Network. Users deposit CNPY and receive cCNPY — a transferable, yield-bearing receipt token — while the underlying stake participates in Canopy committees and earns block rewards. A secondary governance and value-capture token (CLIQ) aligns long-term incentives across users, validators, and the protocol treasury. This whitepaper details canoLiq's architecture, economic model, governance framework, and path to full chain sovereignty within the Canopy incubator model.

1. Introduction & Problem Statement

Staking in Proof-of-Stake networks locks capital: participants sacrifice liquidity for security. In Canopy Network, CNPY must be bonded in committees for validators to earn block rewards and for the ecosystem to be secured. This creates an opportunity cost that deters participation, particularly from DeFi-oriented users who require composable, liquid positions.

canoLiq solves this by abstracting the complexity of committee staking into a single, fungible token (cCNPY). Beyond simple liquid staking, canoLiq leverages Canopy's unique restaking model — where the same bonded CNPY can simultaneously secure multiple Nested Chains — to maximize yield for depositors without requiring additional capital.

Key Value Propositions

- **Liquidity:** cCNPY is freely transferable and DeFi-composable while earning staking yield.
- **Capital Efficiency:** Restaking across multiple Canopy committees compounds rewards on the same collateral.
- **Governance:** CLIQ holders direct protocol parameters, treasury allocation, and validator selection.
- **Canopy-Native:** Built as a Nested Chain — launching with shared security, maturing toward full sovereignty.

1.1 How canoLiq Works: End-to-End

Before diving into technical details, here is a plain-language walkthrough of the full user journey and protocol operation:

Step	Who Acts	What Happens
1. Deposit	User	User sends CNPY to the canoLiq smart contract. The contract mints cCNPY at the current exchange rate and returns it to the user. The user now holds a liquid, yield-bearing position.
2. Delegation	canoLiq Protocol	The pooled CNPY is delegated on-chain to whitelisted professional validator operators registered with canoLiq. These operators have bonded their own CNPY as collateral on the Canopy Root Chain and listed canoLiq's committee ID in their stake configuration.
3. Node Operation	Validator Operators	Each whitelisted validator runs a full instance of the canoLiq Nested Chain software on their own infrastructure — dedicated servers they operate and maintain. They execute NestBFT consensus for the canoLiq chain, producing blocks and submitting Certificate Result Transactions to the Canopy Root Chain each block.
4. Reward Accrual	Canopy Root Chain	Every Root Chain block, new CNPY is minted and sent to the canoLiq committee fund pool (auto-subsidization). Validators divide those rewards per the 70/10/10/10 default split. canoLiq's share accrues in the protocol contract.

Step	Who Acts	What Happens
5. Fee & Distribution	canoLiq Protocol	canoLiq applies its 12% protocol fee to received rewards. 88% flows directly to cCNPY holders via exchange rate appreciation. The 12% fee is split: 40% back to cCNPY holders (extra yield), 30% to DAO treasury, 15% to validator operators, 15% to CLIQ buyback.
6. Redemption	User	User returns cCNPY to the contract. The contract calculates the current exchange rate (original CNPY + accrued yield) and initiates unstaking from the committee. Small redemptions are served from a liquidity buffer instantly; large redemptions follow Canopy's standard unstaking cooldown.

1.2 Who Runs the Validators?

This is the most important operational question for canoLiq. The answer: professional node operator companies, not canoLiq itself.

canoLiq is a protocol, not an infrastructure company. It does not run servers. Instead, it maintains a whitelisted registry of professional validator operators — entities like institutional staking providers who already operate infrastructure on Canopy Network. These operators:

- Register with canoLiq DAO by bonding their own CNPY collateral on the Canopy Root Chain, listing the canoLiq committee ID in their MessageStake transaction.
- Run dedicated server infrastructure with redundant signing keys, uptime monitoring, and slashing protection tooling.
- Accept slashing risk: if they double-sign or go offline beyond the MaxNonSign threshold, their own bonded CNPY is slashed — not users' deposited CNPY directly. The insurance fund compensates users for any shortfall.
- Earn 15% of canoLiq protocol fees as compensation for operational costs and slashing risk exposure.
- Are subject to ongoing performance review and can be ejected from the validator set by CLIQ governance vote if performance degrades.

△ At launch, canoLiq will partner with 5–10 established Canopy validator operators to seed the initial committee. Validator set expansion is governed by CLIQ holders.

2. Canopy Network: Operational Context

Understanding canoLiq requires a firm grasp of Canopy's economic and architectural primitives. Canopy is a recursive blockchain framework in which a Root Chain (CNPY) provides shared security to Nested Chains via opt-in committees of restaking validators.

2.1 NestBFT Consensus

Canopy uses NestBFT, a BFT consensus mechanism fused with Proof-of-Age. Validators organize into Committees — subsets that execute BFT for each Nested Chain independently.

The elected committee leader submits a Certificate Result Transaction to the Root Chain each block, recording rewards and slashing decisions.

2.2 Economic Model

Each Root Chain block, new CNPY is minted and distributed to auto-subsidized committees. A committee qualifies for auto-subsidization when its share of total network restake meets or exceeds the governance-controlled subsidization threshold (currently 33%). This creates a natural TVL competition and incentivizes CNPY holders to actively delegate to growing Nested Chains.

2.3 Reward Distribution (Default)

The default committee reward split for each Nested Chain block is:

Recipient	Share	Description
Block Producer	70%	Root Chain validator who produced the Nested Chain block
Root Chain Delegate	10%	Randomly selected by stake-weight among delegates for this chain
Nested Chain Validator	10%	Stake-weighted random selection on the Nested Chain
Nested Chain Delegate	10%	Stake-weighted random selection on the Nested Chain

△ canoLiq operates on top of these mechanics — it does not modify them. Its fee model captures value from rewards received after Canopy's on-chain distribution.

3. Protocol Architecture

3.1 How canoLiq Sits Inside Canopy

canoLiq is deployed as a Nested Chain on Canopy. As a committee member, it pools CNPY deposits from users and participates in one or more committees. The architecture allows canoLiq to:

- Pool stake from thousands of users into a single, large validator position — lowering the barrier to earning staking rewards.
- Optimize restaking: dynamically allocate stake across committees to maximize yield.
- Abstract complexity: users hold cCNPY with no need to manage committee membership, cooldowns, or validator operations.

3.2 The cCNPY Token

When a user deposits CNPY, canoLiq mints cCNPY at the current exchange rate. The exchange rate increases monotonically as staking rewards accrue to the protocol, meaning cCNPY appreciates relative to CNPY over time (auto-compounding).

Property	Detail
Minting	1:1 at current exchange rate on CNPY deposit
Redemption	At current exchange rate; subject to committee unstake cooldown
Yield Mechanism	Exchange rate appreciation (auto-compounding net rewards)
Transferability	Fully transferable; DeFi-composable
Slashing Risk	Proportional exposure to committee slashing events

3.3 Rewards Flow (Step by Step)

Given a gross reward of X CNPY generated by canoLiq's stake exposure in a given period:

1. Canopy distributes rewards to the committee pool per the 70/10/10/10 split described above.
2. canoLiq receives its proportional share (R) based on its stake weight in the committee.
3. canoLiq applies a 12% protocol fee on R. Fee amount = $0.12 \times R$.
4. The fee is distributed per the canonical 40/30/15/15 split (see Section 4).
5. The remaining 88% of R flows to cCNPY holders via exchange rate appreciation.

⚠ IMPORTANT: Canopy does not apply a protocol-level DAO tax on top of rewards before distribution. All economics flow through committee reward pools as described in Section 2.2. The canoLiq 12% fee is applied to rewards received by the protocol.

4. Protocol Fee Model

4.1 Fee Rate

The default protocol fee is 12% of net rewards received by canoLiq from committee participation. This rate is governance-controlled by CLIQ holders and may be adjusted within bounds set by the protocol constitution (proposed range: 5% – 20%).

4.2 Fee Distribution

Recipient	Share of Fee	Amount (of 12%)	Purpose
cCNPY Holders	40%	4.8% of R	Reinvested into exchange rate; bonus yield on top of base staking APY
canoLiq DAO Treasury	30%	3.6% of R	Protocol operations, grants, security, future development
Validators & Infra	15%	1.8% of R	Operational incentives, uptime bonuses, slashing risk compensation
CLIQ Buyback & Burn	15%	1.8% of R	Open-market CLIQ purchases; default action is burn; DAO may vote quarterly to distribute to locked CLIQ stakers instead

4.3 Effective Yield for cCNPY Holders

cCNPY holders benefit from both the base staking yield (88% of canoLiq's received rewards) and the fee redistribution component (40% of the 12% fee = additional 4.8% of received rewards). The aggregate yield passed through to the exchange rate is therefore $88\% + 4.8\% = 92.8\%$ of received rewards.

5. CLIQ Tokenomics

5.1 Overview

CLIQ is the governance and value-capture token of canoLiq. Fixed supply: 100,000,000 CLIQ. CLIQ is not a reward emission token used to inflate staker yields — its primary roles are governance participation and value capture via buybacks.

5.2 Distribution

Allocation	% of Supply	Tokens	Vesting
Validators & Infrastructure	22%	22,000,000	3-year linear; 12-month cliff
Community & Airdrops	20%	20,000,000	Snapshot-based linear; 12-month distribution
DAO Treasury (canoLiq)	15%	15,000,000	DAO-controlled; time-locked multisig

Allocation	% of Supply	Tokens	Vesting
Liquidity Incentives	15%	15,000,000	Emitted over 24 months; governed by DAO
Founders & Core Team	12%	12,000,000	4-year vesting; 12-month cliff
Strategic Partners	10%	10,000,000	18-month linear; 6-month cliff
Developer Grants & Ecosystem	6%	6,000,000	Milestone-based grants

5.3 Design Rationale

- Validators receive the largest single allocation (22%) because canoLiq's security, uptime, and committee performance directly depend on validator quality. The 12-month cliff (not 6 months) ensures validators remain committed through the highest-risk first year of mainnet operation before any tokens unlock.
- Community & Airdrops (20%) uses a snapshot-based linear emission — eligibility is determined at launch based on Canopy staking history, then distributed daily over 12 months. This is distinct from milestone-based grants (used only for Dev Grants).
- Liquidity Incentives (15%) are sized to support DEX liquidity for cCNPY/CNPY and CLIQ pairs without creating excessive inflation pressure. CNPY subsidies from Canopy DAO (Section 6) are designed to reduce the need to emit all of this allocation early.
- Founders vest over 4 years with a 12-month cliff — longer than typical 3-year schedules — to align team incentives with long-term protocol success.
- Treasury (15%) is held in a time-locked multisig controlled by CLIQ governance, providing a strategic reserve for protocol emergencies, audits, and ecosystem development.

5.4 CLIQ Value Accrual

CLIQ accrues value through three mechanisms:

- Buyback pressure: 15% of all protocol fees are used to purchase CLIQ on the open market. The default action is to burn purchased CLIQ (deflationary). CLIQ holders may vote quarterly to redirect buyback proceeds to locked CLIQ stakers instead of burning.
- Governance premium: CLIQ stakers with locked positions receive boosted voting power and may receive a share of buyback CLIQ rather than it being burned (DAO vote required).
- Protocol ownership: CLIQ holders govern the most critical lever in the protocol — the fee rate — making CLIQ a claim on future protocol cash flows.

6. Canopy Subsidy Strategy

A key pillar of canoLiq's bootstrap economics is the strategic use of Canopy DAO subsidies. To understand why this matters, it helps to understand the problem it solves.

6.1 The Bootstrap Problem

Any new DeFi protocol faces a chicken-and-egg problem: it needs liquidity to be useful, but it needs to be useful to attract liquidity. The standard industry solution is "liquidity mining" — paying early users in the protocol's native token (CLIQ) to deposit funds and provide DEX liquidity. This works, but it has a cost: it floods the market with freshly minted CLIQ before the protocol has proven its value, creating sell pressure that erodes the token price for everyone.

canoLiq's approach is different: use CNPY — subsidized from Canopy DAO — as the bootstrap incentive instead of CLIQ. Early users are still rewarded. But the reward comes in established CNPY, not freshly printed CLIQ. The result is the same user behavior (deposit early, provide liquidity) with far less inflation pressure on CLIQ.

Concrete Example

- Without subsidies: To attract \$5M TVL in month 1, canoLiq might need to emit 500,000 CLIQ as farming rewards (0.5% of supply in month 1 alone).
- With CNPY subsidies: Canopy DAO provides CNPY to seed the committee rewards pool. Early depositors earn bonus CNPY yield on top of their base staking APY. canoLiq emits zero additional CLIQ for the same TVL outcome.
- The 15M CLIQ liquidity allocation then stretches across 24+ months instead of being front-loaded, benefiting all long-term CLIQ holders.

6.2 Auto-Subsidization Mechanics

Canopy's protocol has a built-in subsidy mechanism: every Root Chain block, newly minted CNPY flows automatically to committees that hold $\geq 33\%$ of total network restake (the auto-subsidization threshold). canoLiq's goal is to qualify for this from day one by working with launch validators to reach that stake threshold immediately at mainnet.

6.3 Requesting Canopy DAO Manual Subsidies

In addition to auto-subsidization, any party can manually subsidize a committee via a MessageSubsidy transaction. canoLiq plans to submit a formal proposal to Canopy DAO requesting manual CNPY subsidies during the bootstrap phase:

- Proposed duration: 6–12 months post-mainnet, or until TVL milestones are self-sustaining.
- Proposed tranches: linked to TVL milestones (e.g., CNPY subsidy tranche 1 at \$500K TVL, tranche 2 at \$2M TVL, tranche 3 at \$10M TVL).
- Transparency: all subsidy flows are on-chain and publicly auditable; canoLiq commits to monthly reporting to Canopy DAO.

- Alignment: subsidies requested specifically as bootstrap support, not as permanent operating dependency.

Scenario	CLIQ Emitted (Liquidity Bucket)	Result
No subsidy	15M CLIQ over 24 months (front-loaded)	Heavy early sell pressure on CLIQ
Partial subsidy (6 mo.)	~8–10M CLIQ over 24 months	~30–45% reduction in early CLIQ emission
Full subsidy (12 mo.)	~5–8M CLIQ over 24 months	~50–60% reduction; extended incentive runway

7. Restaking Optimization

Canopy's restaking model allows the same bonded CNPY to simultaneously secure multiple Nested Chains. canoLiq will implement a restaking optimization engine to:

- Identify high-yield committees: monitor reward per CNPY staked across all active Canopy committees and dynamically allocate canoLiq's stake to maximize aggregate APY.
- Manage slashing risk: diversify stake across committees to limit exposure to any single committee's slashing events.
- Balance liquidity: maintain sufficient stake in the canoLiq committee itself to support redemption demand without triggering large cooldown delays.
- Governance-controlled: CLIQ holders vote on restaking allocation policy, including minimum and maximum stake per committee.

8. Governance

8.1 Scope of CLIQ Governance

CLIQ holders govern all critical protocol parameters through the canoLiq DAO:

Parameter	Default	Governance Bounds
Protocol Fee Rate	12%	5% – 20%
Fee Distribution (40/30/15/15)	As specified	Adjustable per component within ranges
Buyback Mechanism	Burn	Burn vs. distribute to locked stakers; DAO vote per quarter
Validator Onboarding	Committee vote	Criteria set by DAO

Parameter	Default	Governance Bounds
Restaking Allocation	Optimization engine	Max/min per committee; DAO-controlled
Subsidy Proposals	Canopy DAO submission	canoLiq DAO approves first
Treasury Spending	Multisig + timelock	Thresholds set by DAO

8.2 Voting Mechanics

- CLIQ can be staked and time-locked for boosted voting power (longer lock = higher multiplier).
- Proposals require a minimum quorum (proposed: 5% of circulating supply) to be valid.
- Emergency proposals (security-critical) may use a 24-hour fast-track with a higher approval threshold (67%).
- All treasury transactions above a governance-defined threshold require a timelock delay (proposed: 48 hours for small, 7 days for large).

9. Risk Framework

9.1 Smart Contract Risk

All canoLiq smart contracts will undergo two independent audits before mainnet deployment, followed by formal verification of core accounting invariants (exchange rate monotonicity, fee distribution correctness). A public bug bounty program will be launched alongside mainnet.

9.2 Slashing Risk

canoLiq's stake is exposed to Canopy committee slashing for double-signing and non-signing. Mitigation:

- Insurance fund: 5% of DAO treasury seeded progressively as a slashing protection reserve, accumulating from protocol fee income over the first 12 months. Target: 5% of peak TVL in reserve within 12 months of mainnet. This fund compensates cCNPY holders for any losses caused by validator slashing events.
- Validator diversification: stake distributed across multiple high-reputation validators with independent infrastructure to ensure no single failure event affects more than a fraction of total TVL.
- Real-time monitoring: automated on-chain alerts for non-signing behavior with rapid governance-triggered ejection from the active validator set, minimizing cascading slashing exposure.
- Slashing cap: Canopy's MaxSlashPerCommittee governance parameter limits the number of slashes per committee per block, providing a protocol-level circuit breaker against catastrophic loss events.

9.3 Liquidity Risk

cCNPY redemptions are subject to Canopy's unstaking cooldown period. canoLiq will maintain a liquidity buffer (CNPY held in reserve) to honor small redemptions without requiring on-chain unstaking, reducing user friction. Large redemptions follow the standard cooldown.

9.4 Concentration Risk

A single protocol controlling a large fraction of Canopy's total stake could pose systemic risks. canoLiq will self-impose a TVL cap of 33% of total Canopy network stake pending ecosystem maturation and governance approval to lift this cap.

9.5 Oracle / Price Risk

The cCNPY/CNPY exchange rate is computed entirely on-chain from protocol state — it does not depend on external price oracles, eliminating oracle manipulation risk for the core yield mechanism.

10. Autonomy Graduation

canoLiq begins life as a Canopy Nested Chain, leveraging shared security. As the protocol matures, it may graduate to a fully sovereign chain (its own Security Root) when it meets objective thresholds:

Graduation Criterion	Target Threshold
Total Value Locked (TVL)	> 50,000,000 USD in CNPY
Active Validators	> 30 independent validators
DAO Participation	> 15% average turnout on governance proposals
On-Chain Activity	> 10,000 transactions/day sustained for 30 days
Treasury Reserves	> 12 months of operational runway

Graduation is subject to cross-chain governance coordination (Canopy DAO + canoLiq DAO) to preserve historical finality and ensure seamless user experience during migration.

11. Roadmap (90-Day Launch Execution)

Phase	Timeline	Deliverables
Foundation	Days 1–30	Smart contract development; fee model implementation (post-

Phase	Timeline	Deliverables
		committee split); CLIQ vesting contracts; testnet deployment
Audit & Subsidy	Days 30–60	Independent audit #1 & #2; Canopy DAO subsidy proposal submission; bug bounty launch; multisig treasury setup
Mainnet Launch	Days 60–90	Mainnet deployment; cCNPY minting live; CLIQ distribution begins; liquidity pools seeded; governance portal live

12. Disclaimer

This whitepaper is a technical and conceptual document provided for informational purposes only. It does not constitute financial, legal, or investment advice. CLIQ and cCNPY tokens involve significant risk, including the loss of principal. Participation in canoLiq should be preceded by independent legal and financial review. The protocol design described herein is subject to change prior to and following mainnet launch based on audit findings, governance decisions, and evolving Canopy Network economics.

canoLiq Protocol — Building Liquid Staking on Canopy Network